2.1. Electric charge & electric field

What comes to your mind when you hear electricity?

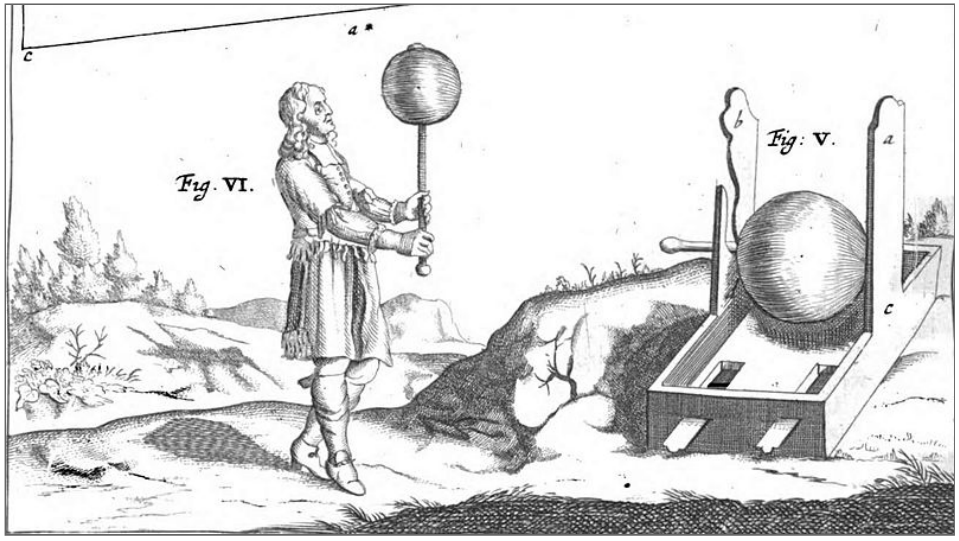
- modern technology, i.e. EVs, computers, smartphones, etc.
- human body relies on electrical signals i.e. heart & brain
- electric forces
    - involved in holding atoms & molecules together
    - are the origin of forces such as friction or normal force can at the atomic level
- gravitational forces remains a separate entity

Short history of studying electricity

- since ancient Greece, the effect of static electricity known
  - “elektron" is Greek for “amber”
  - rub amber with cloth $\rightarrow$ amber attracts small pieces
- scientific investigation of electricity around late 1700:
  - Charles Augustin de Coulomb (1736 - 1806)
  - Benjamin Franklin (1706–1790)
  - Michael Faraday (1791 - 1867)
  - Karl Friedrich Gauss (1777-1855)
  - Alessandro Volta's (1745-1827)
  - ... and many more ...

Otto von Guericke & his sulfur globe

- 1672: reported that after rubbing sulfur globe, globe repealed/attracted light objects
- $\rightarrow$ there is **contactless** force beyond gravitation $\rightarrow$ `es07`



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Static electricity

- are there different types of charge?
- when do they attract and when do they repel each other?
- $\rightarrow$ `es13 + es04`

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Static electricity

- two types of charge: **positive** and **negative**
- arbitrary convention by **Benjamin Franklin** (1706–1790):
    - rubbing amber/plastic → **negative charge**
    - rubbing glass rod → **positive charge**
- charges of **same type repel** each other
- charges of **opposite type attract** each other

Charge origin

- **simplified atomic model:**
  - tiny nucleus with **positively charged protons** and neutral neutrons
  - **negatively charged electrons** orbit nucleus
- **neutral atom:** equal number of protons & electrons
- **ion:** atom that gained or lost electrons
- **polar molecules**: charge **distribution is non-uniform**, e.g. water molecule with positive side (H) and negative side (O)
- **elementary charge**
  - smallest observed charge: **electron or proton**
  - $e = 1.602 \times 10^{-19}\mathrm{C}$

Conductors vs. Insulators

- **free electrons**:
    - electrons that can "detach from their parent atoms" and move through the material/lattice
- **conductors** (e.g. metals): high abundance of free electrons
- **insulators** (e.g. wood): low abundance of free electrons
- **semiconductors** (e.g. silicon): intermediate abundance of free electrons

Conservation of charge

- net electric charge **cannot be created or destroyed**
- $\rightarrow$ **charges can only be separated** but their sum remains constant
- **example**: rubbing a glass rod with a cloth:
    - glass rod gains positive charge by emitting free electrons
    - cloth gains equal negative charge by receiving free electrons
    - **total charge remains zero**
- **grounding**:
    - connection to Earth
    - Earth acts as a **charge reservoir**

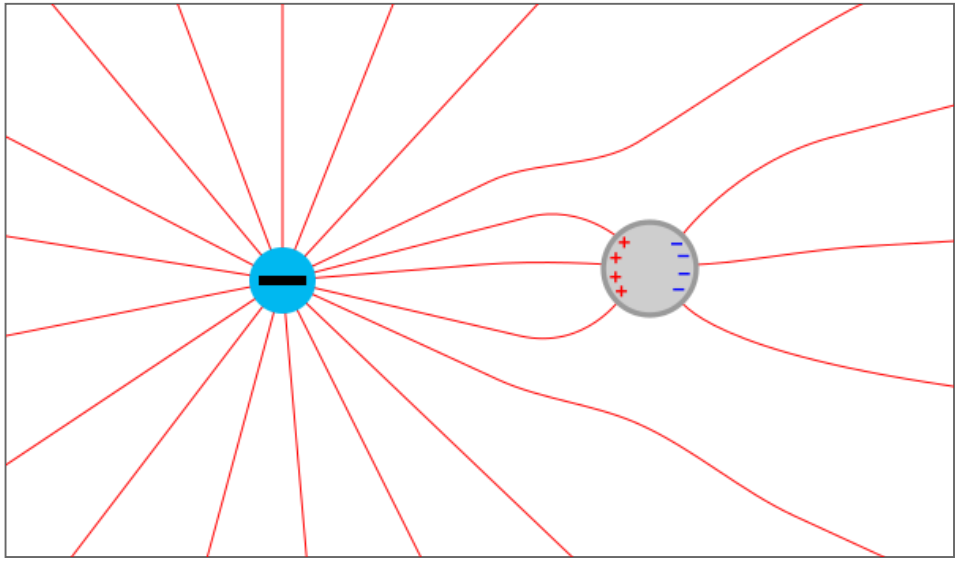
Inducing charge by conduction

`es13 + es04`

- two conductors (one charged, one neutral)
- **contact required**
- electrons redistribute
- both objects end up with the same polarity
- charge distribution depends on geometry

Inducing charge by (electrostatic) induction

- **no contact required**



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Inducing charge by (electrostatic) induction

- **no contact required**
- in conductor/influence: `es10`
    - **redistribution of free electrons** in nearby conductor which causes charge separation
- in insulators/polarization: `es09`
    - electrons cannot move freely
    - **molecules change their orientation** which causes charge separation
- $\rightarrow$ induced charge separation **does not change total charge**

Electrostatic force & Coulomb's law

`es08 + sim Coulomb`

- how to quantify/compute electrostatic force?
- investigated by **Charles Augustin de Coulomb** (1736–1806)
- honoring his contribution, charge Q is measured in the SI unit **coulomb** [C]

Electrostatic force & Coulomb's law

$$F = k\frac{Q_1 Q_2}{r^2}$$

- with:
  - $k \approx 9.0 \times 10^9\ \mathrm{N} \cdot \mathrm{m}^2/\mathrm{C}^2$ (Coulomb's proportionality constant)
  - r is the distance between charges
  - Q_1 and Q_2 are the charges
- **force direction**
  - force acts **along the line connecting both charges**
  - **like charges repel**, **opposite charges attract**

Vector Form of Coulomb's Law

$$\vec{\mathbf{F}}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{Q_1 Q_2}{r^2}\hat{r}_{21}$$

- with:
  - $\vec{\mathbf{F}}_{12}$: force Q_2 exerts on Q_1
  - fundamental constant **permittivity of free space** ε_0
  - $k = \frac{1}{4\pi\varepsilon_0}$
  - $\hat{r}_{21}$ is **unit vector** pointing from Q_2 to Q_1.
- **superposition principle**: for multiple charges, net force is the vector sum of all forces

Concept of the Electric Field

- observations so far:
  - charges q exert electrostatic force $\vec{\mathbf{F}}$
  - electrostatic forces act at a distance & contactless
- **electric field** $\vec{\mathbf{E}}$ connects q and $\vec{\mathbf{F}}$:
  - $\vec{\mathbf{E}}$ is a **vector field**
  - concept by Michael Faraday (1791–1867)
  - analogy with gravitational field: $\vec{\mathbf{F}} = q\vec{\mathbf{E}} \leftrightarrow \vec{\mathbf{F}} = m\vec{\mathbf{g}}$

Electric field form of Coulomb's law

- **idea**: measure field created by charge Q by placing a small *test charge* q (q has negatable effect of Q)
- using Coulomb's law, $\vec{\mathbf{E}}$ is then:

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q} = \frac{1}{4\pi\epsilon_0}\frac{Qq}{r^2}\hat{r}_{21}\frac{1}{q} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\hat{r}_{21}$$

- **superposition principle**: net field is the sum of individual fields:

$$\vec{\mathbf{E}}_{net} = \vec{\mathbf{E}}_A + \vec{\mathbf{E}}_B + \vec{\mathbf{E}}_C + \ldots$$

- **integral form**: integrate over distribution of **infinitesimal small charges** dQ

$$d\vec{\mathbf{E}} = \frac{1}{4\pi\epsilon_0}\frac{dQ}{r^2}\hat{r}$$

$$\vec{\mathbf{E}} = \int d\vec{\mathbf{E}}$$

Field lines of electric field

`sim field lines`

- motivation:
  - $\vec{\mathbf{E}}$ is a vector field, i.e. each point in space is represented by a vector
  - vector length $==$ magnitude of field
  - vector orientation $==$ direction of field
  - as plot, **often perceived as cluttered**
- **concept of field lines**:
  - arrows indicate **direction & magnitude** of $\vec{E}$
  - **density of lines** represents field's magnitude
  - direction of $\vec{\mathbf{E}}$ **tangent** to field lines
  - field lines **start on positive, end on negative** charges

- **field lines do not cross**

Motion of charged particles inside electric field

- a charge q in an **electric field** $\vec{E}$ experiences force:

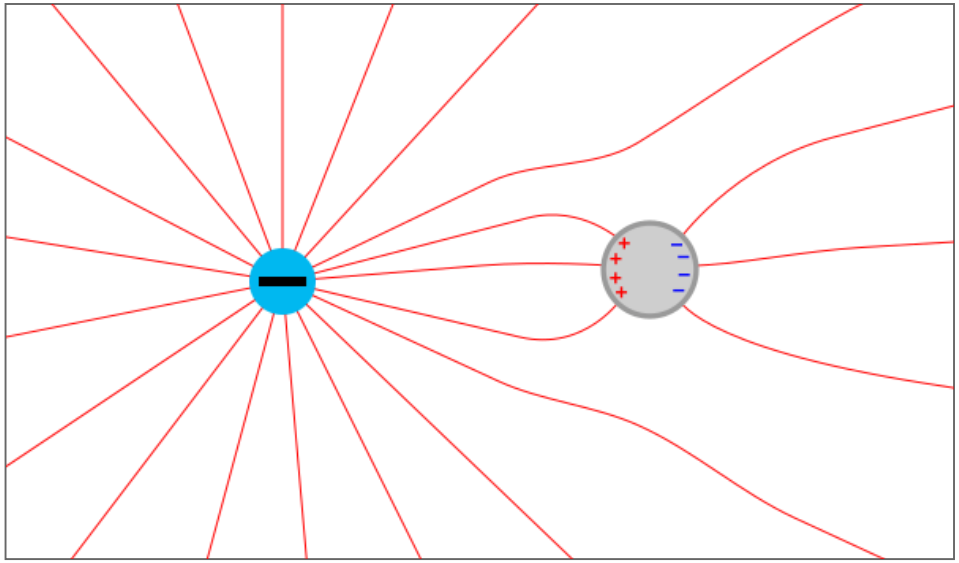
$$\vec{\mathbf{F}} = q\vec{\mathbf{E}}$$

- for an **electron in uniform** E, acceleration is:

$$a = \frac{-eE}{m_e}$$

Conductors in an electric field

- reminder: **electrostatics, i.e. charges at rest**
- for conductors in static condition:
    - **electric field inside conductor is zero**: $\vec{\mathbf{E}} = 0$
    - *if this would not be the case, there would be a force acting on free electrons causing them to move until they a reach position in which the net force is zero, i.e. the field is zero and they are at rest* ($\vec{\mathbf{F}} = q\vec{\mathbf{E}}$)
    - **net charge distributes on surface**
    - electric field **perpendicular** to surface



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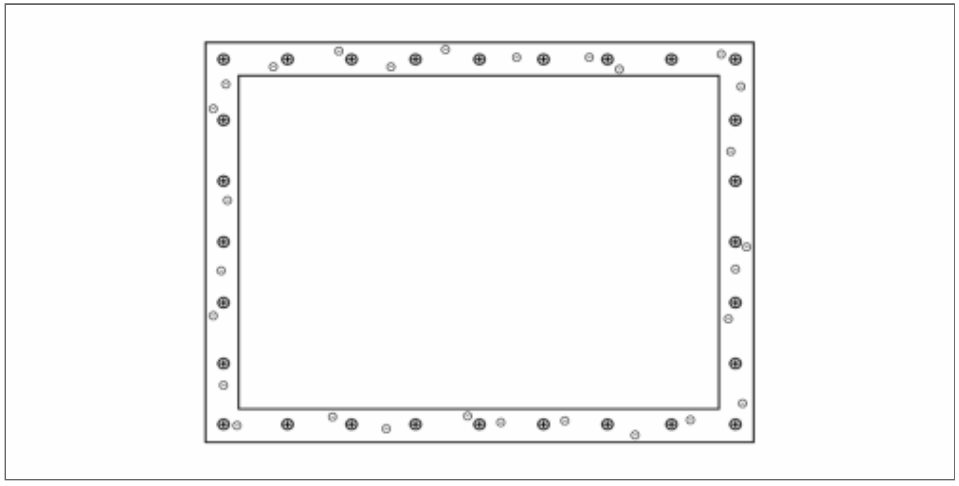
Faraday cage

`es16`

- conducting, closed surface with $\vec{\mathbf{E}} = 0$ inside



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2.1. Electric charge & electric field

- Get ready for journey through electromagnetism in the comming weeks
- Revisit basics & go beyond (Maxwell waiting); waves/optics for the finish
- Stay open-minded & challenge you assumption!

<u>Danger! High Voltage</u>